

D228R 5G Router Module

Public Datasheet | Rev. 1.0

D228R is a compact 5G NR/LTE router module for industrial equipment, in-vehicle systems, remote sites, and OEM integration projects. It combines cellular connectivity, dual Gigabit Ethernet, dual-band Wi-Fi, serial interfaces, configurable IO, and remote management capabilities in a board-level form factor.



Product positioning	Summary
Product type	Industrial 5G router module with expansion interfaces
Target users	OEMs, system integrators, industrial IoT teams, mobile equipment builders
Typical projects	Remote gateway, cellular failover, video backhaul, serial-to-IP, IO-to-MQTT, private-network access
Availability	Evaluation samples and project configurations available by request

1. Key Features

- 5G NR / LTE cellular access with regional module options available by project.
- Dual Gigabit Ethernet for WAN/LAN routing, failover, and industrial gateway deployment.
- Dual-band Wi-Fi for AP, station, relay, bridge, and multi-link access scenarios.

- Three TTL serial ports plus configurable IO channels for industrial data acquisition and control.
- Optional positioning support for GPS/BeiDou data collection and reporting.
- Remote management, status reporting, OTA upgrade, and batch maintenance support.
- Open integration model for firmware defaults, protocol behavior, interface options, and OEM requirements.

2. Typical Applications

Application	How D228R is used
Industrial router module	Embedded inside customer equipment as a cellular router, Ethernet gateway, and remote access node
Mobile video backhaul	Provides cellular, wired, and Wi-Fi link continuity for front-end video equipment
Serial and IO gateway	Connects sensors, controllers, meters, and field devices to MQTT, HTTP, TCP/UDP, or customer platforms
OEM remote operations	Supports remote login, status reporting, OTA upgrade, and platform-based device maintenance

3. Core Specifications

Item	Public datasheet value
Cellular module	Regional 5G NR/LTE module options available by project; reference configuration may use Quectel RM520N-GL module
Cellular standard	3G / 4G / 5G (LTE / NR)
5G bands	NR NSA: n41/n78/n79; NR SA: n1/n28*/n41/n77/n78/n79; final band support depends on selected module and region
4G bands	FDD: B1/B2/B3/B5/B7/B8/B20/B28; TDD: B34/B38/B39/B40/B41
Ethernet	2 x 100/1000 Mbps interfaces, configurable by project as WAN/LAN
Wi-Fi	2.4 GHz 802.11b/g/n 2T2R up to 300 Mbps; 5.8 GHz 802.11a/n/ac 2T2R up to 866 Mbps
Serial / IO	3 x TTL serial ports; 3 configurable IO channels, with I2C multiplexing support by project
SIM	Nano SIM; optional eSIM by project
Power input	7-15 V listed range; recommended operating range and peak current should be confirmed by

	project
Typical power	Less than 5 W listed under full-load reference conditions; peak power depends on network and operating environment
Operating temperature	-40 C to +65 C
Board outline	73.4 mm × 42 mm × 15 mm; verify against mechanical drawing before enclosure design

4. Product Views and Interface Overview

This section provides front and rear product views, interface layout, and mechanical dimension references for appearance identification, interface position confirmation, and preliminary mechanical integration evaluation. Complete pin definitions, electrical thresholds, grounding/shielding design, protection design, connector height, enclosure clearance, and thermal conditions should be further confirmed during the project engineering phase.



Figure 1. Front product view

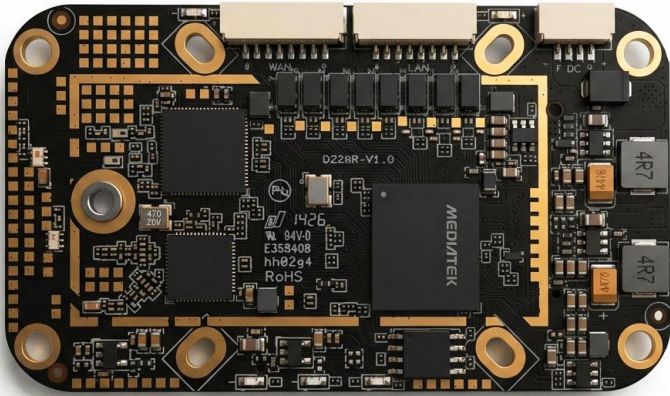


Figure 2. Rear product view

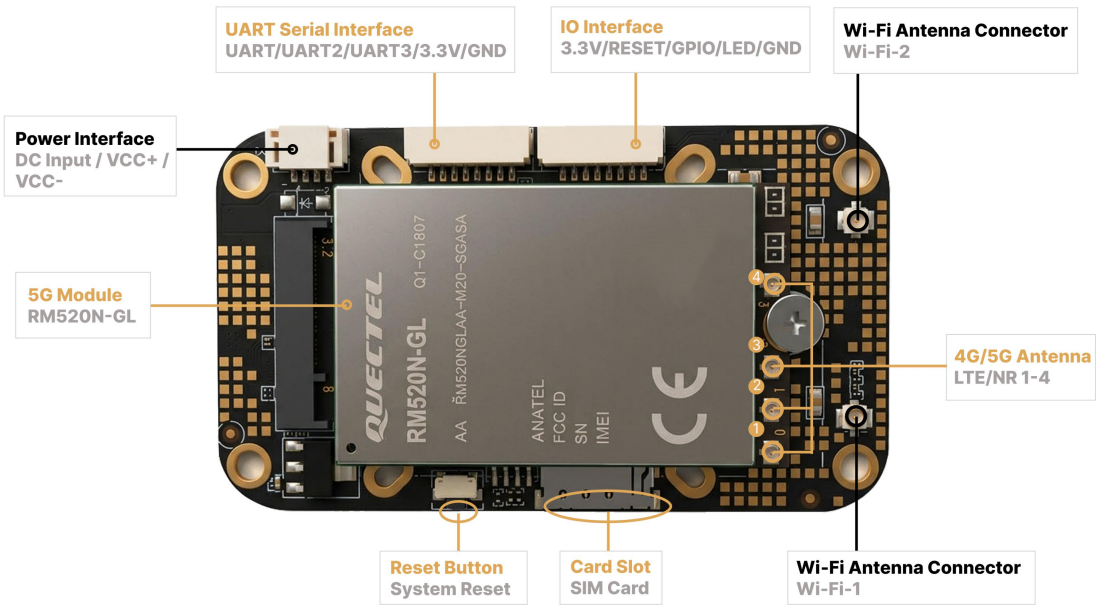


Figure 3. Front interface and major component layout

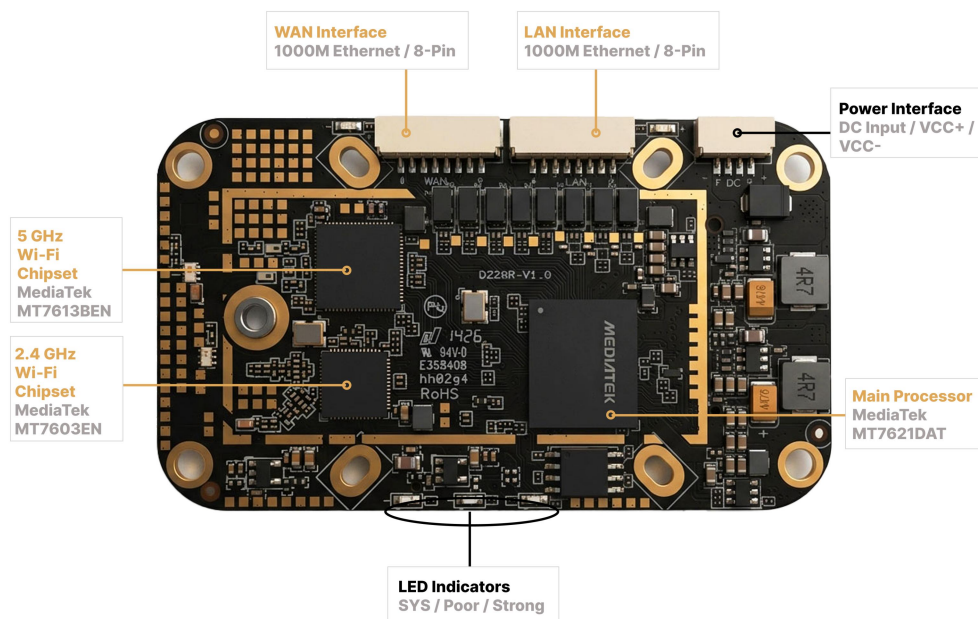


Figure 4. Rear interface and major component layout

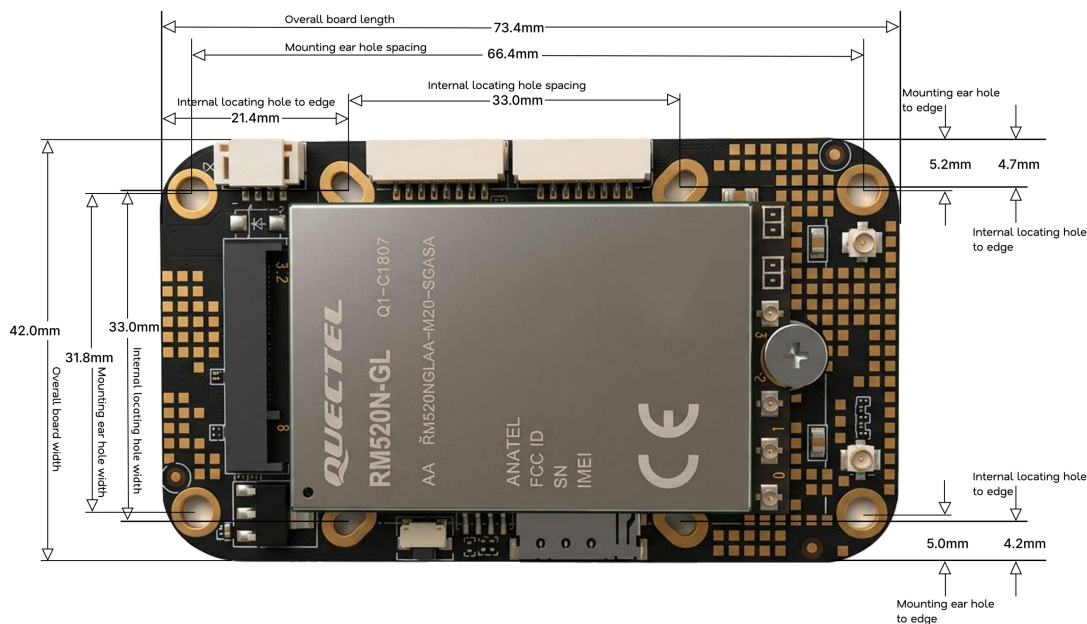


Figure 5. Front mechanical dimensions and mounting reference

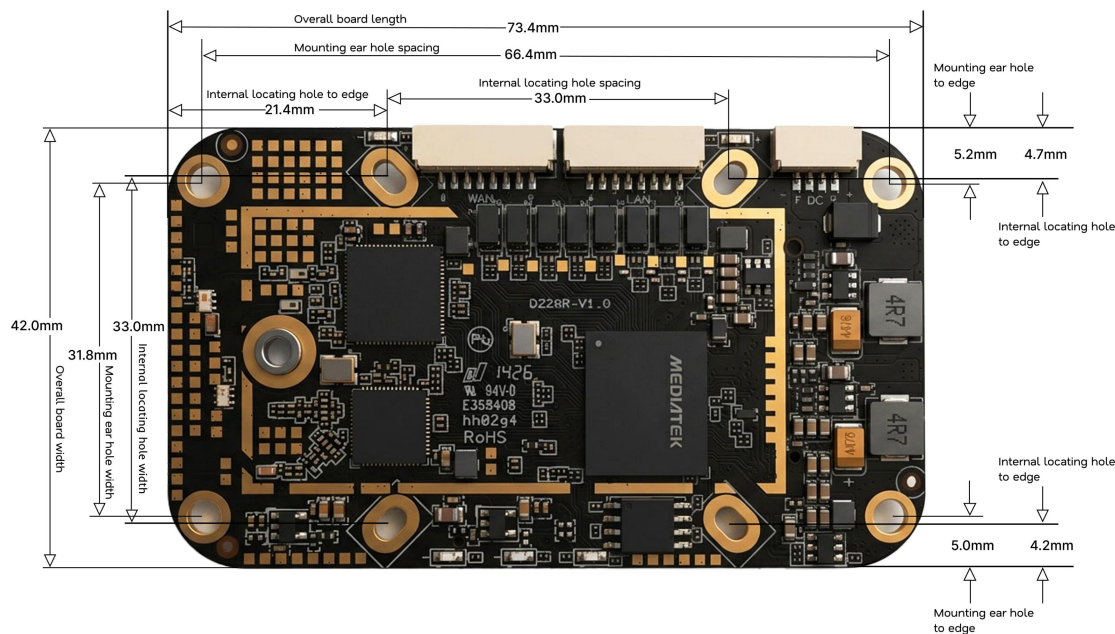


Figure 6. Rear mechanical dimensions and mounting reference

Interface group	Public summary
Power	7-15 V input range is available through the defined board power input; recommended input range, peak current, and protection requirements should be confirmed by project
Ethernet	Two 100/1000 Mbps Ethernet interfaces support WAN/LAN routing, failover, and gateway use; final port behavior depends on firmware and project configuration
Serial / IO	Three TTL serial ports and configurable IO channels support field device connection, data acquisition, protocol conversion, and project-defined control behavior
SIM / Indicators	Nano SIM, system status, cellular signal indicators, and related maintenance signals support field installation and troubleshooting workflows
8-pin expansion interface	The 8-pin interface can be configured on a project basis to support external connection requirements, such as power, serial, I/O, or control signals. Detailed pin definitions are provided during the engineering evaluation phase

Engineering Materials Available on a Project Basis

During the project evaluation phase, detailed pin definitions, electrical thresholds, connector height, component height, DXF/STEP reference files, protection design notes, and enclosure integration evaluation support can be provided as needed.

5. Product Variants

Variant	Description
D228R Standard	Project-selected 5G NR/LTE cellular module unless otherwise specified
D228R-GPS	Adds positioning capability for projects that require location data
D228R Core	For projects that require customer-selected cellular module integration through the onboard M.2 socket; module compatibility, antenna routing, firmware support, and certification scope require project review

6. Software and Remote Management

- Talonbox SkinOS embedded system platform with component-based development and interface extension support.
- Local TCP control, local HTTP status query, LAN discovery, SMS management, and remote HTTP/MQTT reporting.
- Cloud-platform batch management, remote login, remote terminal access, OTA upgrade, and maintenance workflows.
- Firewall, NAT/DMZ, port proxy, static routing, policy routing, SNMP, DDNS, UPnP, and VPN/private networking support.
- Supported VPN and private networking options include PPTP, L2TP, IPsec, GRE, OpenVPN, WireGuard, NAT traversal, and VRRP high availability.

7. Ordering and Project Notes

Item	Public note
Evaluation sample	Available by project request; region, carrier, antenna plan, quantity, and interface requirements should be provided
OEM configuration	Firmware defaults, remote platform settings, branding, protocol behavior, and interface options can be reviewed by project
Lead time	Eligible standard configurations may start from 7 days; final timing depends on quantity, module option, certification, and customization scope

MOQ	Sample and production MOQ should be confirmed by project
Supporting files	Mechanical drawing, interface layout, and engineering package are available during project review

8. Certification and Configuration Notice

- Certification status, cellular module options, antenna configuration, supported frequency bands, product labels, reference images, and delivery timing may vary by target market and project configuration.
- Module-level certification marks shown in reference images apply to the cellular module only. They do not automatically indicate final D228R board-level certification or end-product certification.
- The final customer product, including enclosure, antenna design, power input, installation method, and regional deployment requirements, may require additional engineering review or certification.

9. Contact and Project Review

To request a datasheet package, evaluation sample, OEM integration review, mechanical files, or remote management support, please contact Talonbox with your project requirements.

Recommended information to include

Target region, carrier requirements, expected quantity, enclosure plan, installation environment, required interfaces, and whether the project requires a standard D228R configuration or an OEM integration review.

- Email: talonbox@ashyelf.com
- Website: <https://www.talonbox.com>
- Inquiry form: <https://www.talonbox.com/#contact>

Final configuration, delivery timing, certification scope, and engineering file availability are confirmed by project.